User authentication based on smartphone application usage patterns through learning classifier systems

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***Abstract*—Smartphones have become more ubiquitous than ever. People are installing various applications on their smartphone to fit into their lifestyle. Existing research shows that there are patterns within the ways people access those applications, whether it involves particular locations, particular ranges of time, or many other factors. In this research, through a collaboration with a commercial company, we collected usage data from a popular smartphone application that gives its users access to digital flyers information for shops and supermarkets throughout Japan. Our early experiments found that the pattern information contained inside the data could be used to authenticate users. In this research, we are proposing a behavioral authentication model implementing customized learning classifier systems to search through vast amount of possible patterns to authenticate users of the application. Our early findings for this ongoing research demonstrate that our model can feasibly be a good alternative for additional authentication factor to implicitly authenticate users beyond the initial registration.**

***Index Terms*—Behavioral Authentication, Learning Classifier System, Genetic Algorithm**

I. Introduction

Smart devices such as smartphones have become increasingly essential for many people's daily routines. The convenience brought by many smartphone applications has changed the way people spend their time and do their activities. It is now common for them, for example, to check shops inventory or order food through the relevant applications instead of going to the actual shops. With increased use of those applications, more and more potentially sensitive private data, such as addresses or credit card numbers, are being collected from users. The nature of personal data being gathered by the applications led users to give more care in securing their devices.

Knowledge-based authentications such as passwords and PIN (Personal Identifying Numbers) codes are popular approaches to authenticate smartphone users [1], but they are easy to forget and prone to be re-used by users due to the inconvenience of needing to remember multiple passwords or codes [2]. Possession-based authentications [3] such as devices with built-in hardware token or software token avoids the necessity of remembering passwords, but they also add another layer of inconvenience by requiring users to possess something that can be lost and stolen [4].

Meanwhile, biological biometrics-based authentications [5] avoid the inconveniences brought by both knowledge-based and possession-based authentications. Biological biometrics authentication, as the name suggests, requires users to provide the authentication system with biological data which they already have and carry all the time, such as their fingerprints or facial expression. While biological biometrics-based authentications offer the most seamless authentication method, research has shown that, with effort, it is possible to replicate biological data and use them to fool the authentication system [6].

For even stronger authentications, combinations of two or more of the above authentication methods can be implemented [7]. This type of approach is commonly referred as Multi-Factor Authentication (MFA) [8]. A typical example of MFA is the process of having to input a code that is sent to a user's phone after they correctly entered their password to log-in into a web service. The inconvenience of MFA adds up as more authentications being required [9].

All the previously mentioned approaches typically authenticate users at the initial log-in or unlock session, and nothing prevents other people from using the authenticated account if the account owner happens to forget to log out or to lock the device. While it is common for a service to require re-authentication when a certain amount of time has passed or when important information needs to be changed, the period between those authentications is still prone to misuse [10].

To address this issue, it is necessary for systems to continuously authenticate users and methods offering such approach are often known as continuous [11] or implicit [12] authentication methods. Since knowledge-based, possession-based, and biological biometrics-based authentication procedures are impractical for this kind of situation because they require constant user interaction, behavioral traits are more suitable because they are easy to monitor, smooth to process, and hard to fake [13].

Behavioral authentications are convenient because authentication process can be continuously done implicitly and unobtrusively in the background without interrupting users' activities. This paper is focusing on this particular authentication method. We designed our proposed method with smartphone users in mind, where thefts, may happen when their smartphone is still in unlocked status and thus may provide unauthorized people to access personal applications.

This research is focusing on one particular behavioral trait, which is application access behavior. Regularly accessed application represent patterns within people behaviors and in this paper we are proposing a method implementing Genetic Algorithms (GA) to recognize such patterns and to correctly authenticate people that match the patterns.

Through a collaborative research project, called MITHRA

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Project [14], between Social ICT Research Center of The University of Tokyo and various commercial companies, we managed to gather a collection of behavioral data from over 50,000 smartphone users. To evaluate the feasibility of our proposed method, we conducted early experiments on a small subset of the data from the users of a popular smartphone application in Japan called Shufoo, which is provided Toppan Printing Co., Ltd. The application gives access to its users to view and manage digital flyers from various shops and stores around Japan to acquire information such as the latest price or discount on products being sold there. Our early results from experimenting on the data demonstrate that our model can be a good alternative for additional authentication factor to implicitly authenticate users beyond the initial registration.

II. Related Work

Various research on behavioral authentications has shown that behavioral patterns could potentially be used to distinguish and authenticate users. Fridman, et al. [15] conducted an experiment to authenticate users on mobile devices through processing various data gathered from stylometry information, application usage, web browsing, and Global Positioning System (GPS) location data. Their research is motivated by a survey that showed a large percentage of smartphone users do not bother to lock their phone. They proposed an authentication method that actively monitors users' interaction with their device to secure it. Their findings showed promising potentials for behavioral authentications through smartphone sensors, but suffer limitations from users whose activities did not generate enough data from particular sensors.

Hugo, et al. [16] proposed support vector machines classifiers to continuously authenticate users on mobile devices by analyzing their typing motion behavior. They demonstrated a relatively high accuracy through reliable true and false positive rates. However, their implementation is limited only to applications which require frequent typings and may not be suitable for other kinds of applications.

Meanwhile, Khan and Urs [17] proposed application-centric implicit authentication methods for smartphones. These authentications were designed with different kinds of applications, such as launcher, browser, map, and comic applications. They utilized users' behaviors in terms of number of touch points and swipes to distinguish users. In the paper, they reported a false acceptance rate between 5 to 20 percent for various types of applications. They concluded their research was promising but still needed further tuning to achieve higher accuracy.

More recently, Yamaguchi et al. [18] conducted a large-scale behavioral authentication through multi-modal classification algorithms. They designed their authentication mechanism based on a dataset containing information about list of applications installed on smartphones, as well as the frequency and periods the users were accessing those apps. They discovered that users tend to access certain applications at particular range of days and time. They successfully classified patterns within those users' access behaviors and further shows promising results can be achieved from learning user behaviors. Similarly,

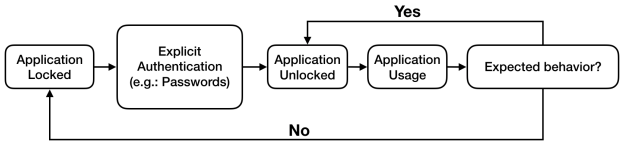

Fig. 1: Illustration of behavioral authentication flow

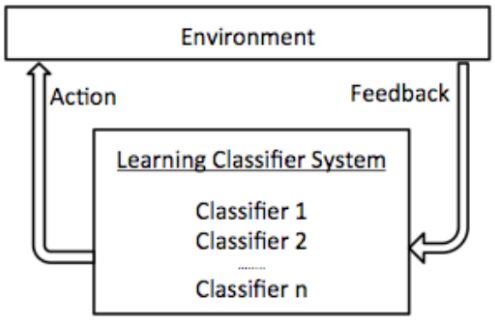

Fig. 2: Learning classifier systems (LCS) model

Ashibani and Mahmoud [19] also conducted a multi-feature authentication based on interactions with mobile applications. Their proposed method demonstrated a relatively high accuracy after analyzing access events with various applications on users' smartphones.

Yamaguchi [18] and Ashibani [19] research is different from our research in a way that their authentication method was designed with device-level authentication in mind where device system could collect behavioral data from multiple applications, and our authentication method was designed with application-level authentication in mind where authentication is required to access a particular application based on that application's usage data. Their promising results on authentication through access events encourage us to design a behavioral authentication that smartphone application developers can implement without requiring access to device-level system.

III. Proposed Approach

In this paper we are proposing a behavioral authentication method which can be used to implicitly authenticate access to an application through learning from a user's past behaviors while interacting with the application (figure 1). As long as the system does not detect unexpected behaviors, users are continuously authenticated while using the application.

A key challenge in behavioral authentication is that people's behaviors changed, and authentication system has to adapt to those changes. A particular machine learning model, called Learning Classifier System (LCS) [21] (figure 2), has been found to be good at adapting to users changing situations [22]. It is essentially a collection of Genetic Algorithms (GA) [23] classifiers that evolves according to feedbacks from a Reinforcement Learning (RL) [24] environment. In our case, GA is useful to evolve LCS classifiers to better adapt when users' usage patterns begin to change.

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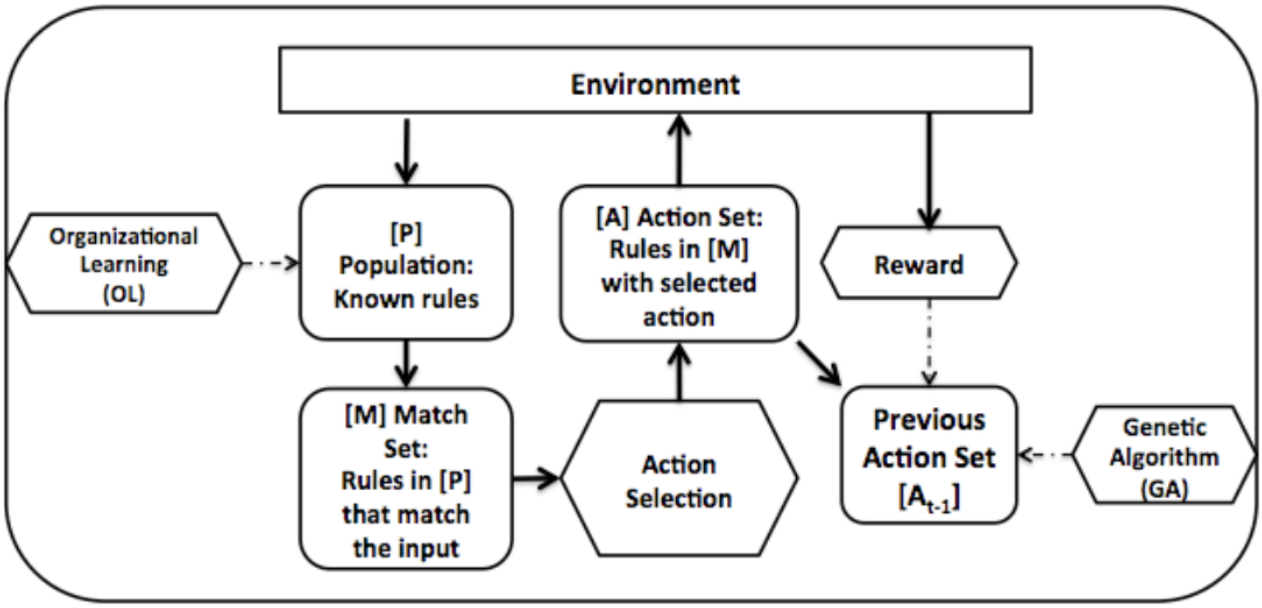

Fig. 3: Architecture of organizational-learning-based LCS

For this research, we are implementing a custom LCS model, called XOLCS [25] (figure 3), which is specifically customized to adapt to smart device users' changing preferences through additional layer of organizational learning (OL) parameters to GA classifiers.

LCS is a collection of condition-action classifiers that represent an intelligent agent's knowledge from previously learnt experience. The system receives a string of input from an environment that it interact with, informing agents about current situation such as the information about which page is being accessed at certain time. After gathering these inputs, agents perform an authentication process and propose actions to either accept or reject further access. LCS allocates reward back to the classifiers that proposed the action, essentially informing them the accuracy of their action and applies genetic algorithms to find and test actions on evolved classifiers.

The condition part of a classifier is an input string received from the application. In our work, the classifier is represented as a string of information of a user access in form of:

| User ID | Page ID | Day | Time |

The action part of a classifier is the choice of action that the classifier proposes. An example of a condition-action classifier would be:

| User ID | Page ID | Day | Time | -> | Accept |

The above classifier can be read as: "if this user is trying to access this page on this day and this time, accept the access". Additionally, each classifier maintains several properties attached to them:

- Prediction: reflects the reward that the classifier anticipate to get when the proposed action is executed given the condition is satisfied.
- Accuracy: reflects the probability the classifier actually getting the predicted reward.
- Prediction error: reflects the difference between the predicted reward and the actual reward.
- Fitness: reflects the relative accuracy of the classifier to other classifiers which propose the same action.

Classifiers inside the population represent the situation that the agent has encountered in the past. The ultimate goal of LCS is to maintain a collection of classifiers with high accuracy inside the population. There is a limit to the size of population allocated to LCS. OL parameters of XOLCS help LCS maintain the classifiers that reflect users' changing behaviors. They maintain several OL parameters to separate rules to be kept from rules that may be "forgotten" or overwritten by new rules when the agent's memory is full. The memory is considered full when the capacity for the number of classifiers an LCS can keep has reached its maximum. OL parameters ensures that when such situation happens, classifiers with the most outdated behaviors are the ones being deleted. As such, the classifier collection grows according to the most recent users' behaviors.

Finally, our proposed method assumes that the communication between smartphones and service providers to be secure with no risk of leak. Furthermore, the method also assumes that data collected during the training phase to be authentic and not spoofed.

IV. Experiment

We conducted our experiments by training our proposed method with data representing access behaviors from smartphone application users. The data contains information about

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obfuscated user ID, access date and time, as well as various information about applications being used.

A. Dataset

The dataset for our experiments came from a collaborative project between Social ICT Research Center of The University of Tokyo and Toppan Printing Co., Ltd., which provided usage data from users of a popular smartphone application called Shufoo. The application gives access to its users to view and manage digital flyers from various shops and stores around Japan to acquire information such as the latest price or discount on products being sold at the stores. This data is part of a bigger project called MITHRA project [14], which collected several behavioral data, including GPS location history and application usage history, from more than 50,000 participants during a period of 3 months in year 2017. More details about the complete MITHRA dataset can be found on [15]. During the data collection period, users of the applications were presented with an option to participate in a university project that analyze smartphone users' lifestyle habit. A webpage address of the project on the university website is also linked. By participating in this project, users agree to continuously share information about their smartphone device, application usage, as well as their location data detected by smartphone's sensor. Users were clearly informed that the collected data would be used only for analysis by the university laboratory and privacy would be strictly protected. The project passed an extensive review by the university's ethical committee and deemed to be appropriately implemented.

We conducted an experiment to test the accuracy of our proposed approach with Shufoo dataset gathered through MITHRA project. Since in this paper we are focusing on measuring the feasibility and accuracy of our proposed method, scalability is considered as future research. Thus, in this paper we are not using the whole collection of Shufoo data, but rather focusing on a small random subset of users.

B. Experiment and Result

We conducted early experiments for users with various number of interactions with Shufoo application. We differentiate between groups of users from less than 10 total interactions up to 100 total number of interactions, where each group consists of 30 users. We trained our custom LCS model with the first two months of access data contained in the dataset. Each user maintain their own LCS classifier population, and the classifiers were exclusively trained on their assigned user. This prevent the requirement to gather all users data into a central place and provide the ability to run the authentication locally on the device. Table 1 summarize the LCS parameters with the value commonly used in other research [22][25]. Figure 4 illustrate the average False Accept Rate (FAR) and False Reject Rate (FRR) for each group. As shown by the figure, result from users with less than 50 times of interaction did not generate high enough accuracy for behavioral authentication. On the other hand, interaction number from 70 times and above reliably produce FAR below 5% and FRR below10%. These results translate into an average of at least one interaction per day would be enough generate those FAR and FRR.

TABLE I: LCS parameter values for the experiment

Parameter	Value
Minimum reward	0.0
Maximum reward	10.0
GA frequency	25
Crossover probability	0.8
Mutation probability	0.05
Wildcard probability	0.3
Learning rate	0.2
Discount factor	0.7
Minimum error threshold	0.1
Fall-off factor	0.1
OL frequency	3
Experience threshold	10

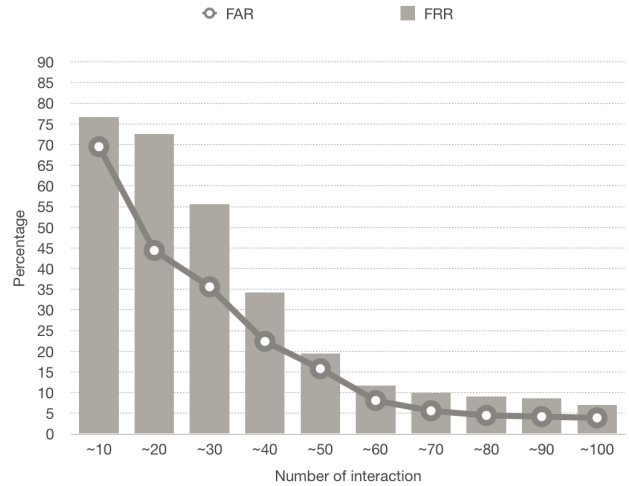

Fig. 4: FAR and FRR for various number of interaction events

Figure 5 summarizes the average FAR and FRR for the group the produce the best result shown in figure 4. It shows the FAR and FRR values when they are averaged according to different number of users. Since each user maintain their own LCS population and the classifiers inside are independent from other user behavior, the average FAR and FRR are relatively similar regardless of the total number of users being compared. These results show that accuracy is improved with more interactions from a user, and it is not affected by the number of users itself. Consequently, as a point of discussion, this discovery led us to believe that our model can be suitable to be implemented as a local on-device behavioral authentication. The reason is because it does not need data from other users to be gathered together at a central server. Usage data, as well as other potentially sensitive data, can be kept locally within smart devices and the device can simply request a copy of LCS learning model from a server to be trained offline on the device itself. This potential approach drives LCS into better privacy and can raise users' trust towards service providers.

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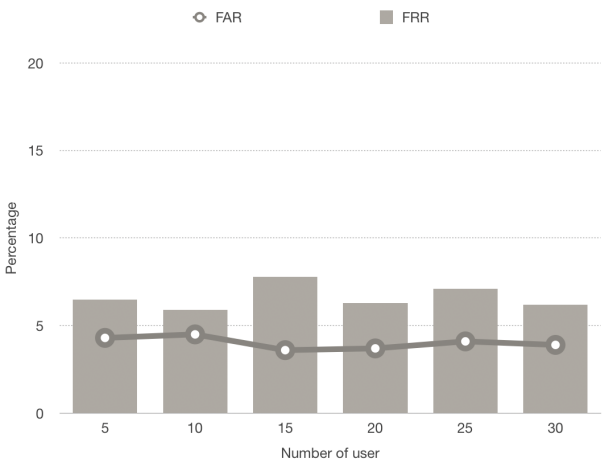

Fig. 5: FAR and FRR for various number of users with up to 100 interaction events

V. Conclusion

Although the results were promising, behavioral authentication is still not perfect to be used as a replacement for traditional authentication method. Our model requires at least two months of regular access history before being able to reliably authenticate, thus the period from before the third month is prone to mistake. However, beyond those limit, we believe our behavioral authentication method is good enough to be implemented as an additional factor after the first explicit authentication to constantly monitor for unexpected behavior to secure access to an application.

Our research is still in its early phase. We are planning to test the scalability of our proposed method by training the authentication system with the whole collection of dataset. Optimization of the algorithm to further increase the accuracy is also considered. Our discovery of potential applicability of our model as a local on-device authentication also led us to further designed our authentication model for Internet of Things (IoT) and smart homes which emphasize the necessity to keep behavioral data as close as possible to the source. This is important as nowadays more and more people are having concerns about sharing application usage with service providers. Thus, we are aiming to improve our model to work with less data leading to shorter training time.

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